Case Study :- Design a real-time add click aggregator

https://github.com/veena-rani/ad-click-analytics/blob/main/README.md

Also a detailed high level design for this in excalidraw.

This GitHub link contains the code for one of the functionalities of the Ad Click Analytics system. The repository includes a README.md file with instructions on how to run the application, the tech stack used, and other important information.

Here I am sharing a few of the running glimpse to highlight the running code .

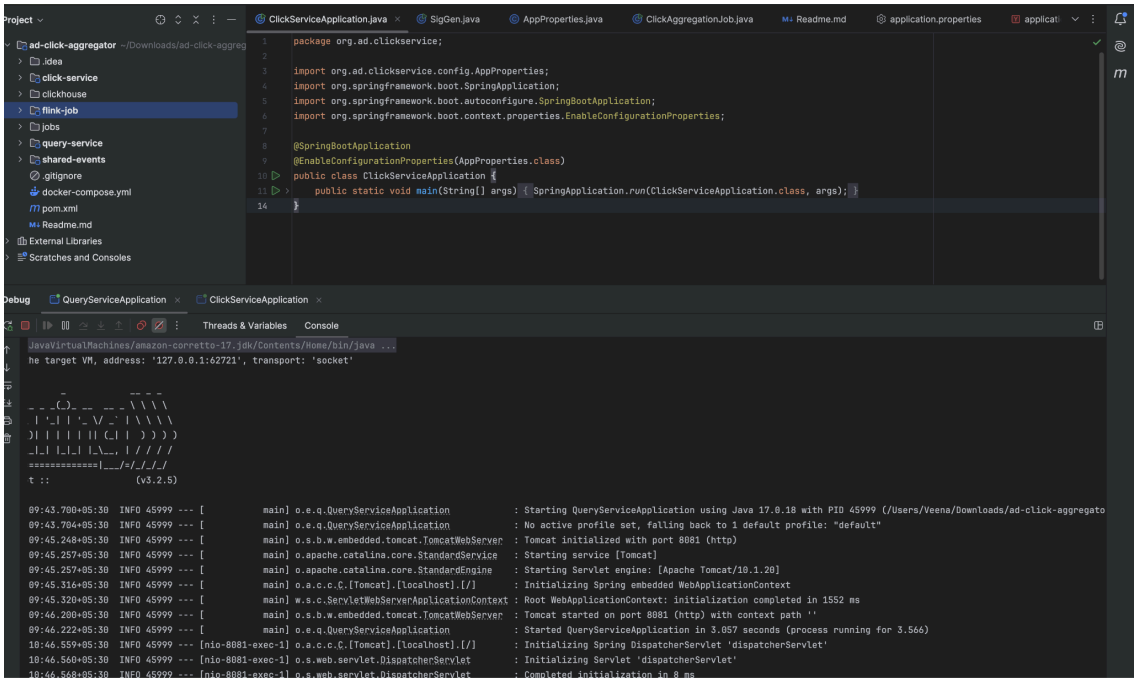
These are the UI which i have introduced for better understanding .

- ClickHouse UI at **http://localhost:8085**
- ClickHouse HTTP at **http://localhost:8123**
- Flink UI at **http://localhost:8082**
- Kafka UI at **http://localhost:8084**
- Redis UI at **http://localhost:8083**

First we need to do docker compose to build the container of all dependency . Then

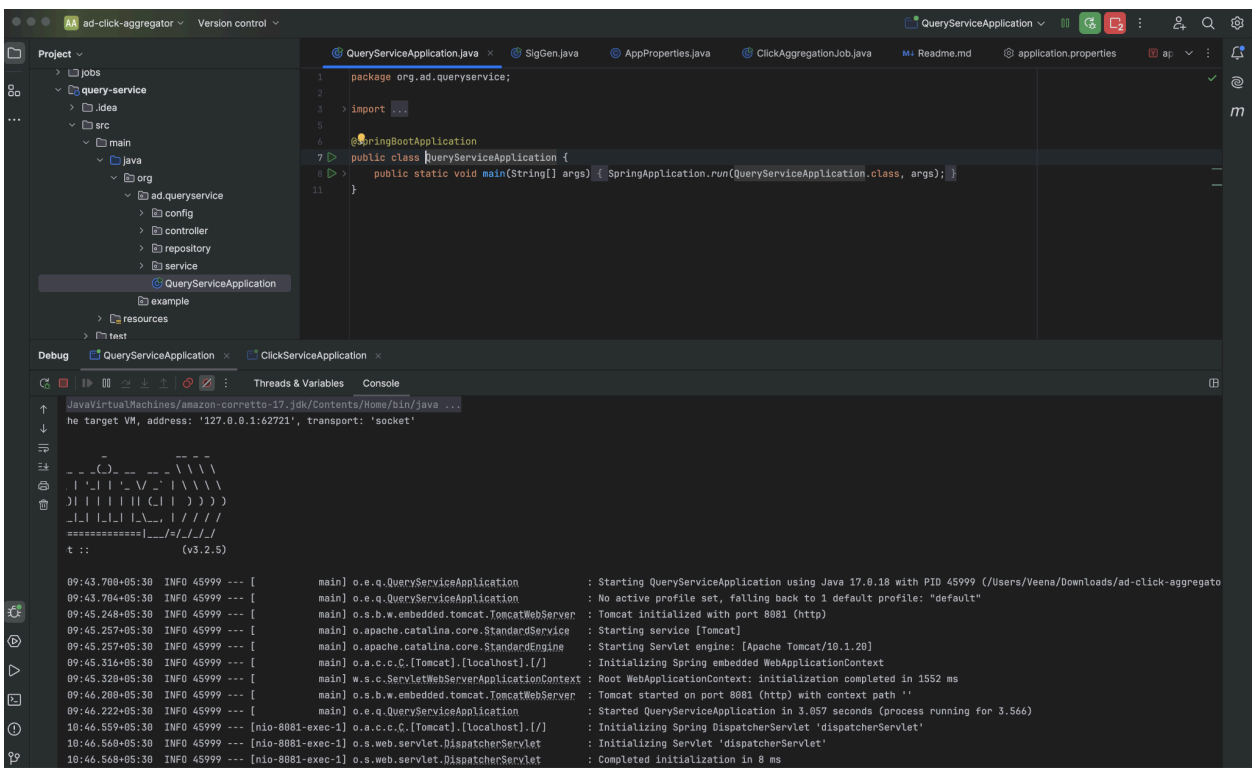
Step 1

You can start with the starting up your backend application click-service



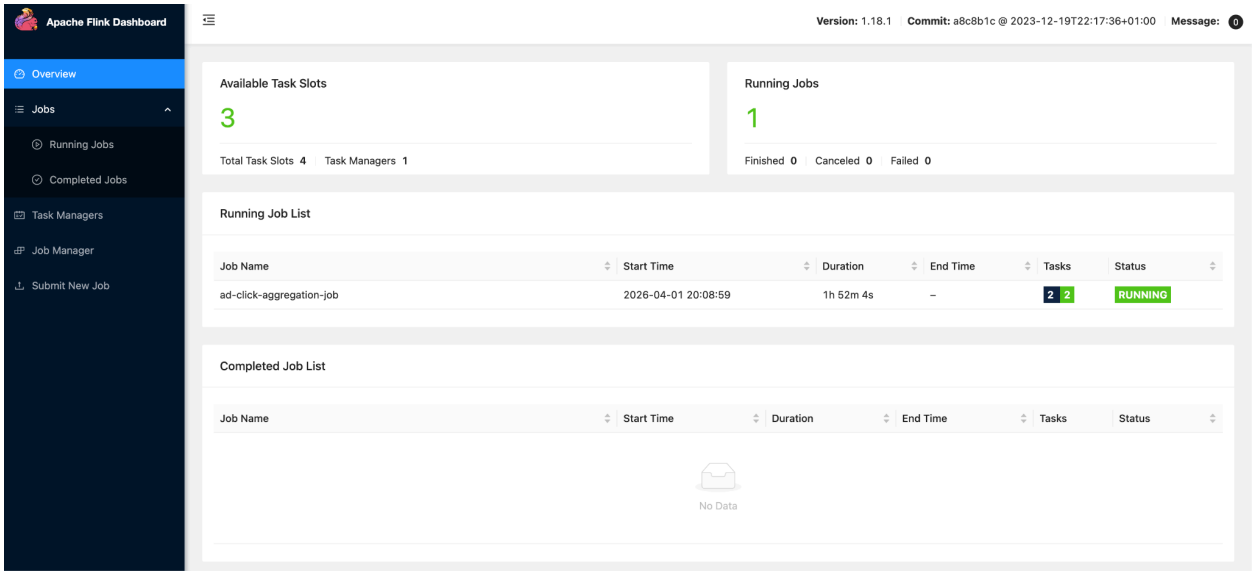
Step 2

Then you can start your query service.



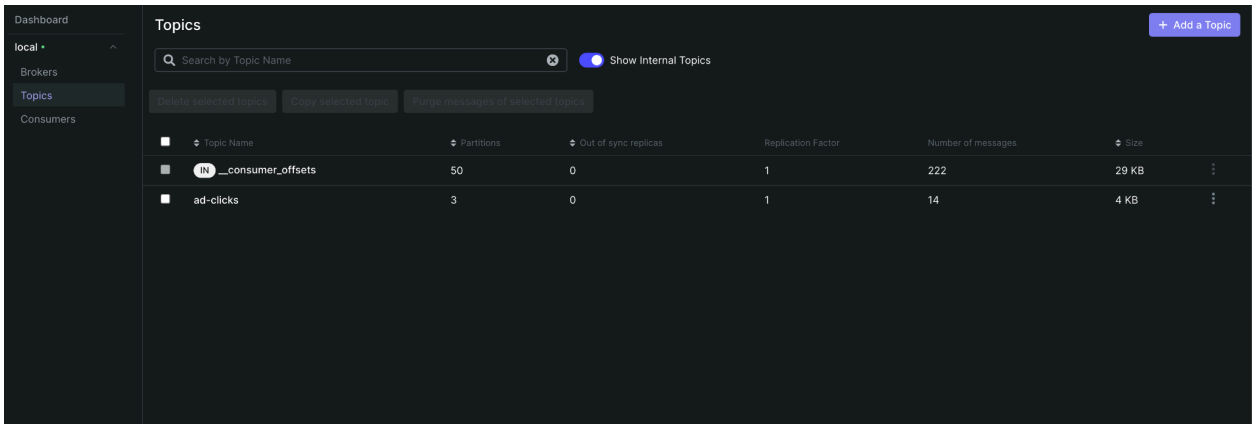
STEP-3

Now you can check your kafka flink and clickhouse that if everything is running or not



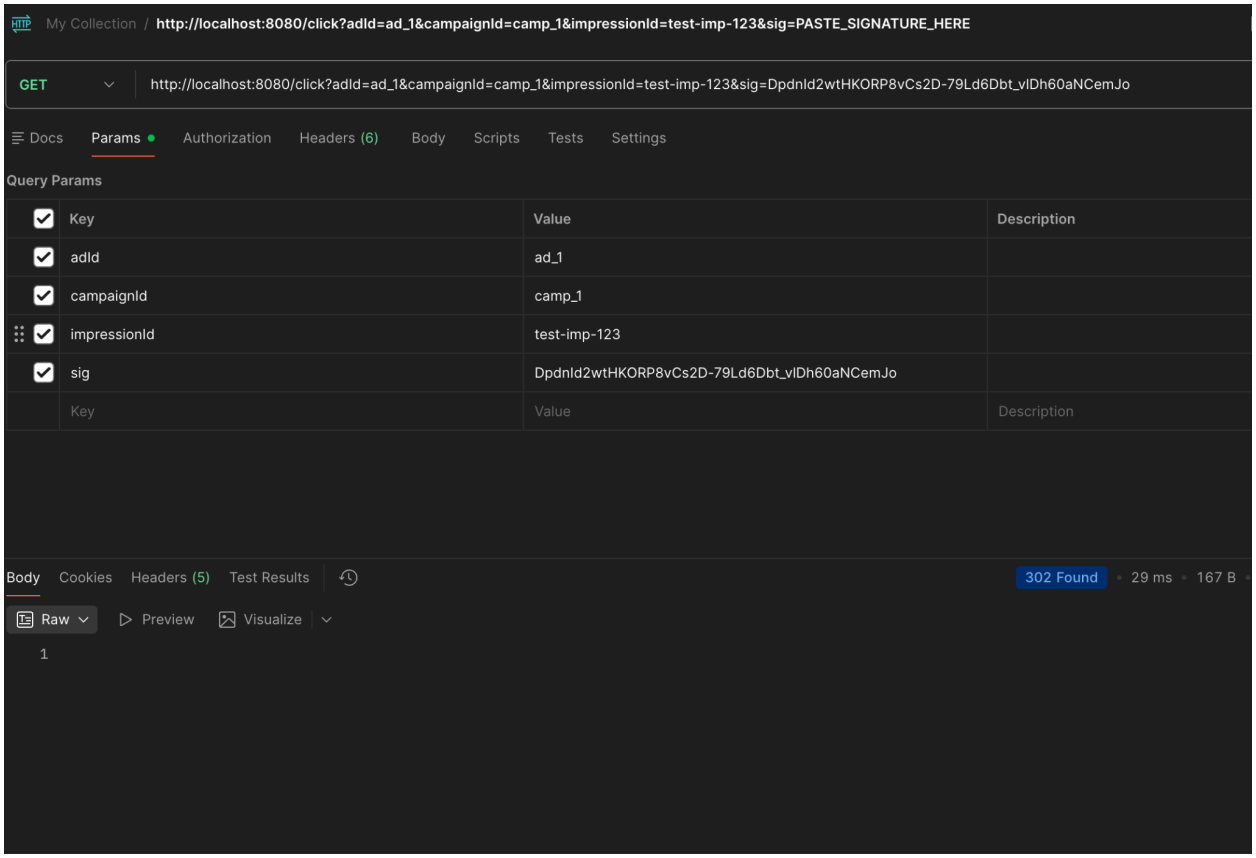
Step-4

Then you can check kafka console



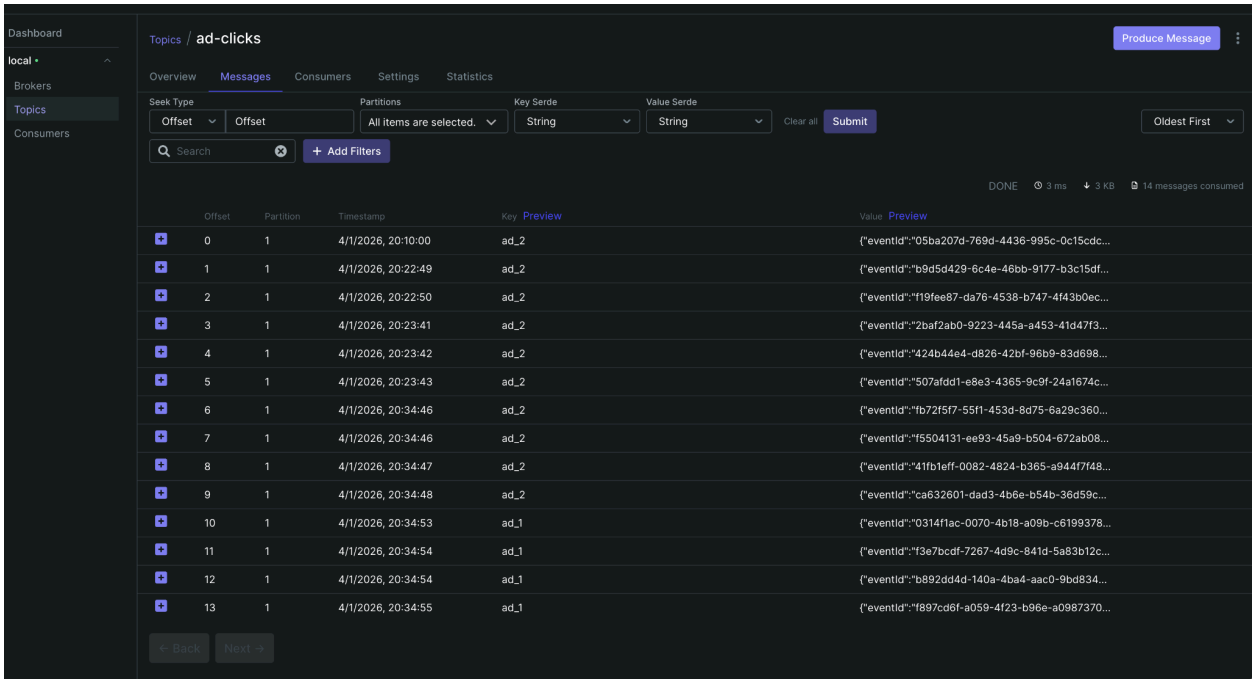
Step- 5

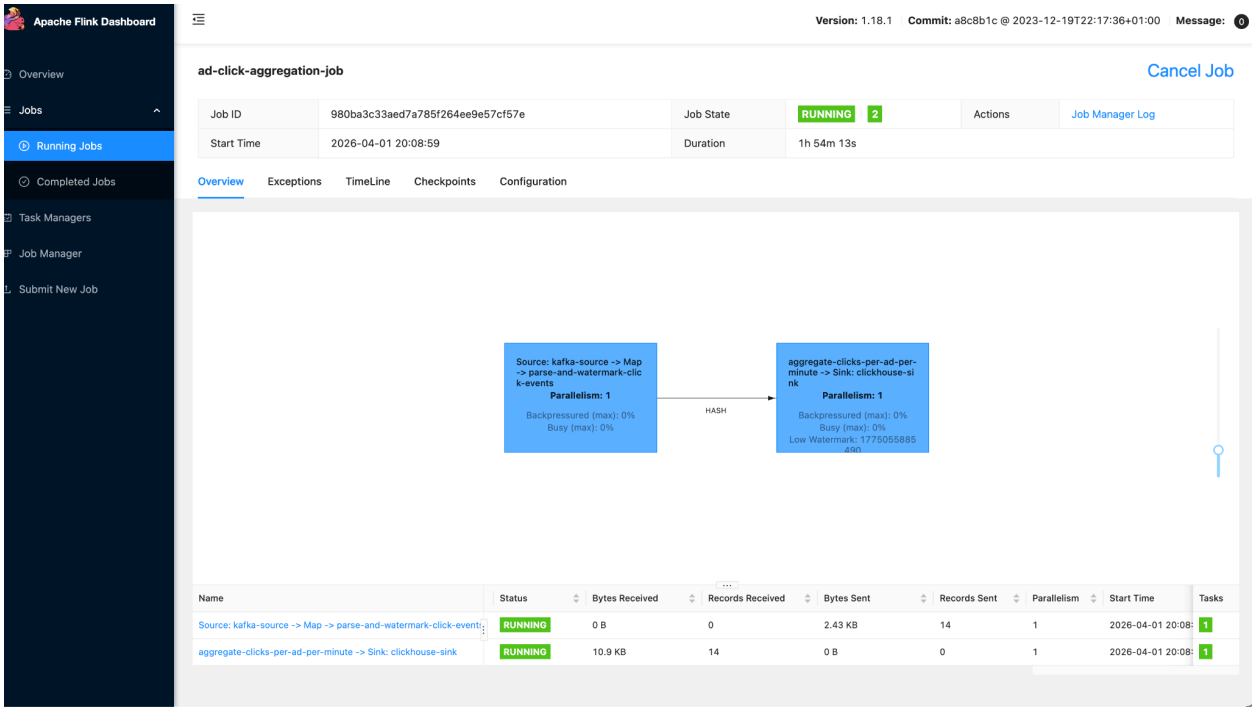
Now once everything is up you can send a click request from postman.



You will receive a successful response with a 302 redirect. Then, you can check the Apache Flink and Kafka dashboards to verify that the message has been published and that the Flink job is running and sending the event to the ClickHouse database.

Step -6





In Flink, you can see the number of records received and records sent, as well as the source and destination metrics. If you are facing any issues, you can check the logs either in the Docker console or in the Flink UI.

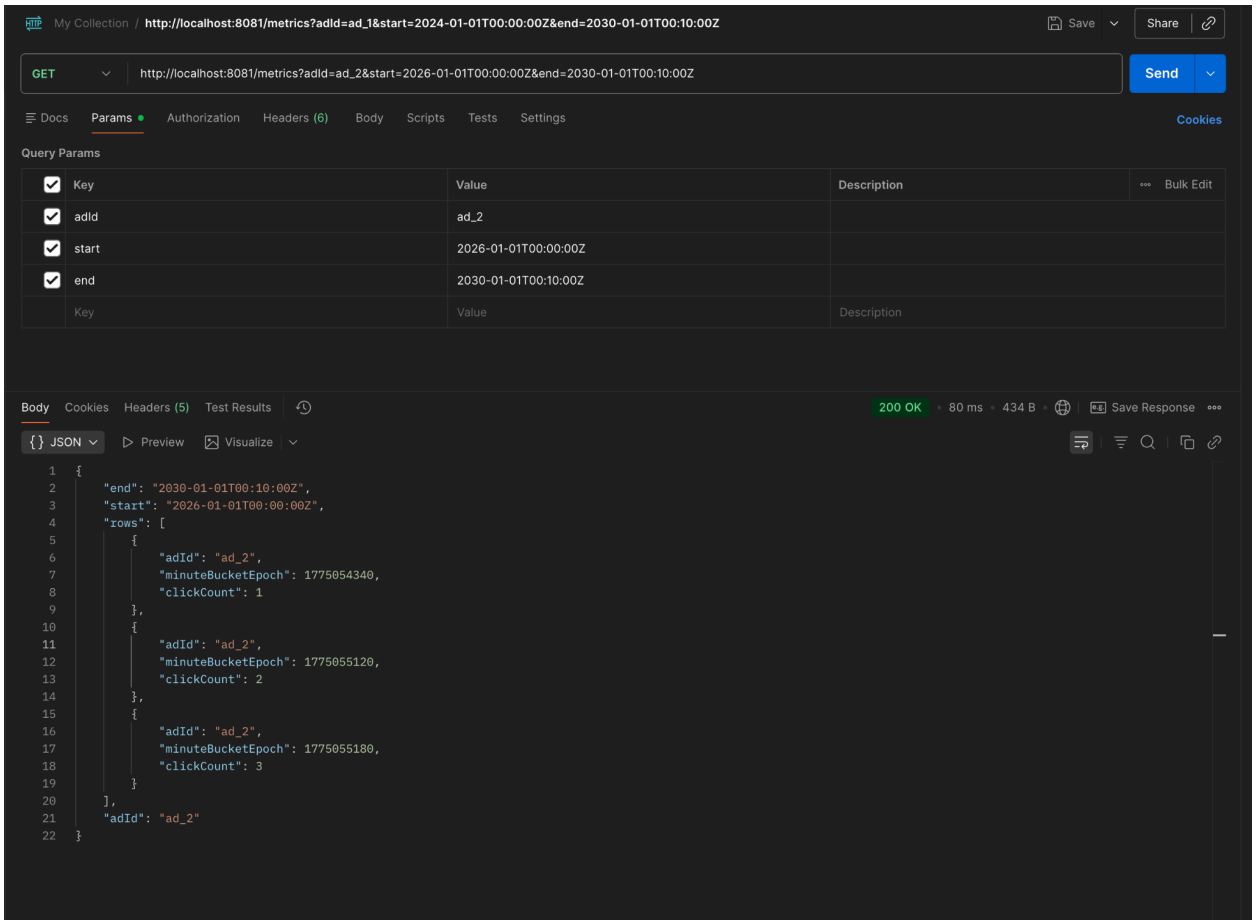
Step-7

You can also check in the clickhouse if the message has been processed to DB or not

```
SELECT *
FROM ad_click_metrics_minute

Query id: 4cd0eb7a-73ac-4ad3-adfb-de507d607e94

   ┌─ad_id─┬─────minute_bucket─┬─click_count─┐
1. │ ad_2  │ 2026-04-01 14:39:00 │           1 │
   └───────┴─────────────────────┴─────────────┘

   ┌─ad_id─┬─────minute_bucket─┬─click_count─┐
2. │ ad_2  │ 2026-04-01 14:52:00 │           2 │
   └───────┴─────────────────────┴─────────────┘

2 rows in set. Elapsed: 0.007 sec.

944a62966f47 :) select * from ad_click_metrics_minute

SELECT *
FROM ad_click_metrics_minute

Query id: a69818f4-6043-4ac5-ac13-5237360c5a82

   ┌─ad_id─┬─────minute_bucket─┬─click_count─┐
1. │ ad_2  │ 2026-04-01 14:52:00 │           2 │
   └───────┴─────────────────────┴─────────────┘

   ┌─ad_id─┬─────minute_bucket─┬─click_count─┐
2. │ ad_2  │ 2026-04-01 14:39:00 │           1 │
   └───────┴─────────────────────┴─────────────┘

2 rows in set. Elapsed: 0.005 sec.

944a62966f47 :) select * from ad_click_metrics_minute
```

Step-8

You can hit the analytics query service to fetch the counts of the click per add



Thank you
Happy learning :-)